

Digital SOIL MOISTURE METER

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A soil moisture meter is used for indicating the water content of a given soil sample. As crop production requires water at different stages and in different amounts, it is important to measure soil moisture from time to time to know its status.

The circuit presented here, for monitoring the soil moisture content in irrigation farms, is an electronic device that measures soil moisture content accurately and precisely. It measures the resistance of flow of the electric current between two

metallic probes. These probes act as sensor elements that register moisture and change it into an electric value. This value is further processed into information in the form of an electronic display.

Circuit and working

Front-end of the soil moisture meter is a simple soil moisture sensor head, built around a few easily-available parts. Working of the sensor head electronics is self-explanatory, so just follow the conceptual drawing (and schematic drawing) shown in Fig. 1 to proceed with its construction.

Complete circuit diagram of the soil moisture meter is shown in Fig. 2. After construction of the sensor head, power it off a regulated 5V DC supply.

Insert the probes of the sensor head into the soil to determine its moisture content, and carefully adjust the preset VR1 so that voltage just above 3V is available at pin A1

of Arduino (Board1) if the soil is wet enough.

Repeat the calibration process with randomly-collected soil samples ranging from very dry to very wet, depicting different moisture levels. Re-tune VR1, if necessary, until you are satisfied with the calibration process. (Note that the depths of penetration of the probes



Fig. 3: Live LCD screen in author's prototype

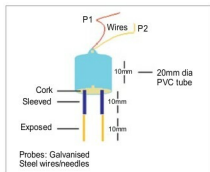


Fig. 1: Soil moisture sensor head

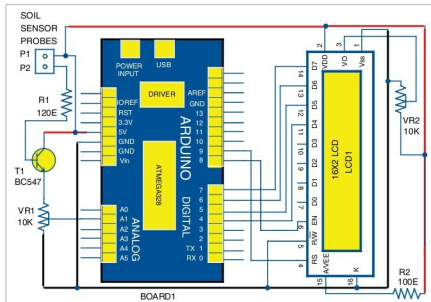


Fig. 2: Circuit diagram of the soil moisture meter

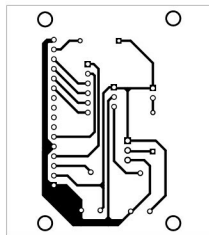


Fig. 4: Actual-size PCB pattern of the soil moisture meter

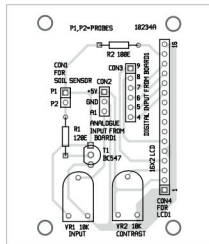


Fig. 5: Component layout of the PCB